

Thomas Jiang

thomas.jiang@duke.edu

tsjiang02.github.io

Department of Mathematics,

Duke University,

120 Science Drive,

Durham, NC 27708-0320

Research Interests

Differential geometry, geometric analysis (PDE, geometric measure theory, Riemannian convergence theory), special holonomy metrics (G_2 , $\text{Spin}(7)$, Calabi-Yau, hyper-Kähler, etc.), gauge theory.

Education

Duke University

PhD, 2023--

Advisor: Mark Haskins

Stony Brook University

Honors B.S., Mathematics, GPA: 3.97/4.00, *summa cum laude*, 2020-2023

Thesis: "The Kummer Construction of Ricci-flat Kähler Metrics on $K3$ via Weighted Analysis", url:

tsjiang02.github.io/papers.html

Advisor: Simon Donaldson

Research Experience

Current work: constructing new complete non-compact torsion-free G_2 manifolds with full holonomy G_2 .

Past work: reproved a conjecture of physicists Page and Gibbons and Pope on the construction of Ricci-flat Kähler metrics on the $K3$ surface using modern weighted analysis/singular perturbation theory.

Publications

Preprints

- P1. Jiang, T. *The Kummer Construction of Calabi-Yau and Hyper-Kähler Metrics on the $K3$ Surface, and Large Families of Volume Non-collapsed Limiting Compact Hyper-Kähler Orbifolds* 2024. arXiv: [2501.08491](https://arxiv.org/abs/2501.08491) [math.DG].

Talks

Geometric Analysis Learning Seminar, November 9th 2022

"Connections with L^p -bounds on curvature"

Awards & Honors

Michael and Jane Fasullo Scholarship in Math

Stony Brook University College of Arts and Sciences, 2021-2022

Kuga-Sah Memorial Award: Freshman (math department award for best freshman)

Stony Brook University Department of Mathematics 2020-2021

Kuga-Sah Memorial Award: Sophomore (math department award for best sophomore)

Stony Brook University Department of Mathematics 2021-2022

Stony Brook Foundation Award for Excellence in Mathematics

(math department award for best undergraduate)

Stony Brook University Department of Mathematics 2022-2023

Conferences Attended

SLMath Geometry and analysis of special structures on manifolds, November 18th-22nd, 2024

SLMath summer school on geometric analysis, June 16-29th, 2024

SCSHGAP meeting, May 13-17th, 2024

Simons/IMPA workshop, March 11-15th, 2024

SCSHGAP meeting, September 7-13th, 2023

PCMI 2019 Graduate Summer School, June 30-July 20th, 2019

Languages and Skills

English (native), Mandarin (conversational)

LaTeX

Last updated: February 15, 2026